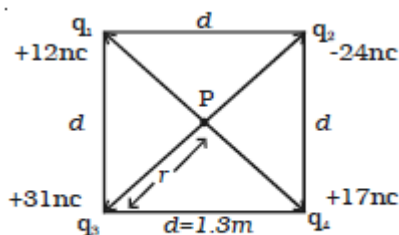


Practice problem sheet Course: PHY 2105 Fall: 2023

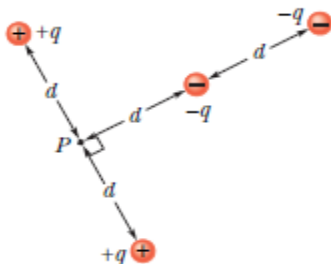
Title: Physics Content: Electric Potential

- Four charges $+2q$, $+4q$, $+2q$ and $-2q$ are placed at the corners of a square. (i) Draw the arrangement of the charges (ii) Calculate the net electric potential at the intersection of the diagonals of the square of side 10 cm if $q = 57 \times 10^{-9} \text{ C}$.
- Calculate (i) the potential at a point due a charge of $4 \times 10^{-7} \text{ C}$ located at 0.09 m away (ii) work done in bringing a charge of $2 \times 10^{-9} \text{ C}$ from infinity to the point.
- Calculate the electric potential at a point P, located at the centre of the square of point charges shown in

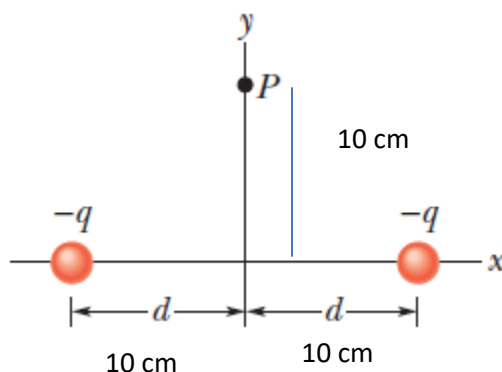


the figure.

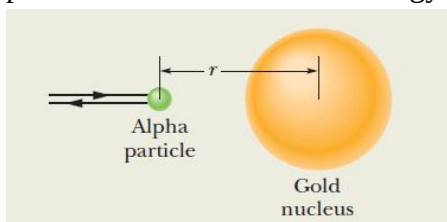
- In Fig. below, what is the net electric potential at point P due to the four particles if $V = 0$ at infinity, $q = 5.00 \text{ fC}$, and $d = 4.00 \text{ cm}$?



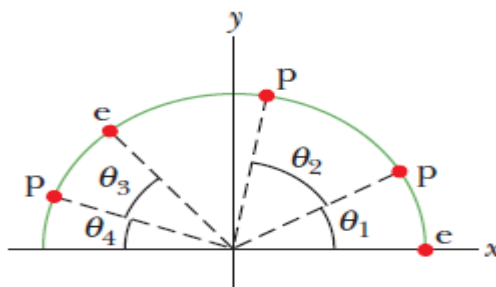
- The electric potential difference between the ground and a cloud in a particular thunderstorm is $1.2 \times 10^9 \text{ V}$. In the unit electron-volts, what is the magnitude of the change in the electric potential energy of an electron that moves between the ground and the cloud?
- Two equal charges of $10 \times 10^{-5} \text{ C}$ are shown in fig below; each produces an electric field at point P on Y axis. (a) What is the magnitude of the electric potential at P?



7. A charged particle produces an electric field with a magnitude of 5.0 N/C at a point that is 60 cm away from the particle. What is the magnitude of the particle's electric potential and charge?
8. Fair weather atmospheric electricity 100 N/C is acting downward 100 km high in the ionosphere. What is the ionosphere voltage required?
9. The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47 D, where $1\text{D} = 1 \text{ Debye unit} = 3.34 \times 10^{-30} \text{ Cm}$. Calculate the electric potential due to an ammonia molecule at a point 52.0 nm away along the axis of the dipole. (Set $V = 0$ at infinity)
10. A proton (mass m_p , charge $+e$) and an alpha particle (mass $4m_p$, charge $+2e$) approach one another with the same initial speed $v = 2 \times 10^8 \text{ m/s}$ from an initially large distance. How close will these two particles get to one another before turning around?
11. An alpha particle (two protons, two neutrons) moves into a stationary gold atom (79 protons, 118 neutrons), passing through the electron region that surrounds the gold nucleus like a shell and headed directly toward the nucleus (Fig. below). The alpha particle slows until it momentarily stops when its center is at radial distance $r = 9.23 \text{ fm}$ from the nuclear center. Then it moves back along its incoming path. What was the kinetic energy K_i of the alpha particle when it was initially far away?

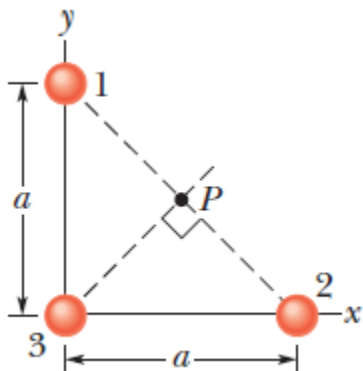


12. A charge of $-1.0 \mu\text{C}$ is located on the y-axis at the coordinates (0,1) while a second charge of $+3.0 \mu\text{C}$ is located on the x-axis at the coordinates (3,0) from the origin. Determine the value of the following quantities at the origin (i) the electric potential (assuming the potential is zero at infinite distance), and (ii) the energy needed to bring a $+3.0 \mu\text{C}$ charge to this position from infinitely far away.
13. Figure shows an uneven arrangement of electrons (e) and protons (p) on a circular arc of radius $r = 2.00 \text{ cm}$, with angles $\theta_1 = 30^\circ$, $\theta_2 = 50^\circ$, $\theta_3 = 30^\circ$, and $\theta_4 = 20^\circ$. What is the magnitude of the net electric

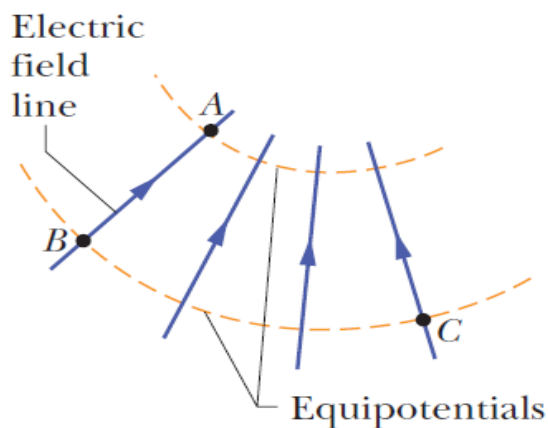


potential produced at the center of the arc?

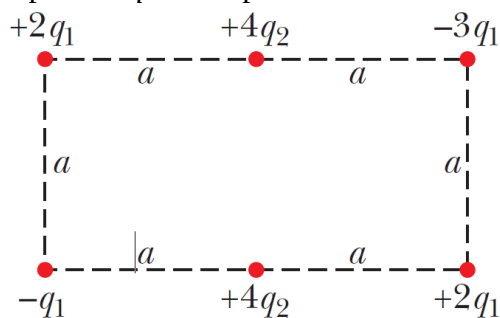
14. The three particles are fixed in place and have charges $q_1 = q_2 = +e$ and $q_3 = +2e$. Distance $a = 6.00 \mu\text{m}$. What are the magnitude of the net electric potential at point P due to the particles?



15. When two electron moves from A to B along an electric field line in below figure, the electric field does 3.94×10^{-19} J of work on it. What are the electric potential differences (i) $V_B - V_A$, (ii) $V_C - V_A$, and (iii) $V_C - V_B$?



16. Figure below shows a rectangular array of charged particles fixed in place, with distance $a = 39.0$ cm and the charges shown as integer multiples of $q_1 = 3.40$ pC and $q_2 = 6.00$ pC. With $V = 0$ at infinity, what



is the net electric potential at the rectangle's center?

17. In the figure below, how much work must we do to bring a particle, of charge $Q = +16e$ and initially at rest, along the dashed line from infinity to the indicated point near two fixed particles of charges $q_1 = +4e$ and $q_2 = -q_1/3$? Distance $d = 1.40$ cm, $\theta_1 = 43^\circ$, and $\theta_2 = 60^\circ$.

